

ETS052

Internet Routing

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Routing

- The Concept of Routing
- Unicast Routing
- Multicast Routing (quick overview)

Figure 19.3 Network layer in an internetwork

Network Layer / Layer 3

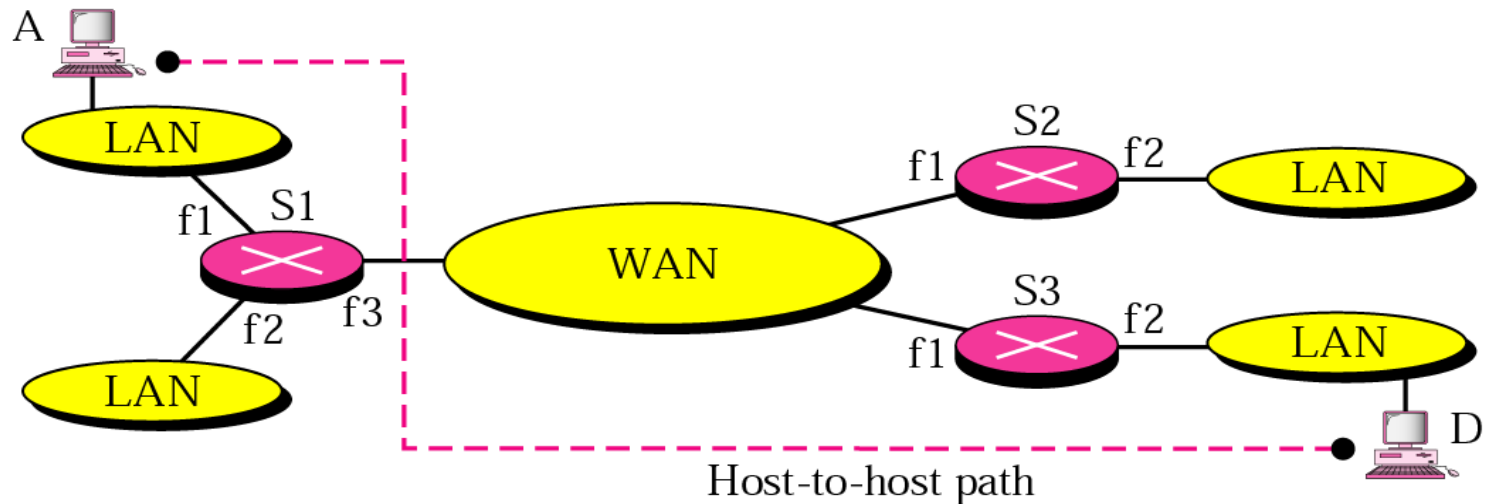
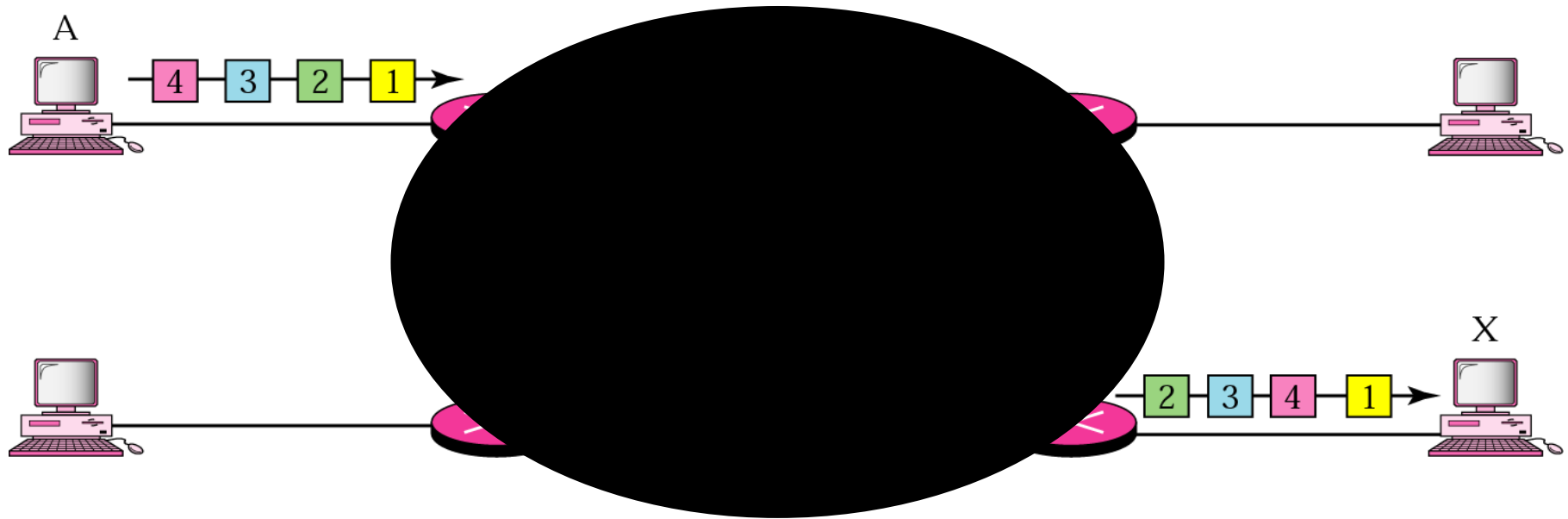
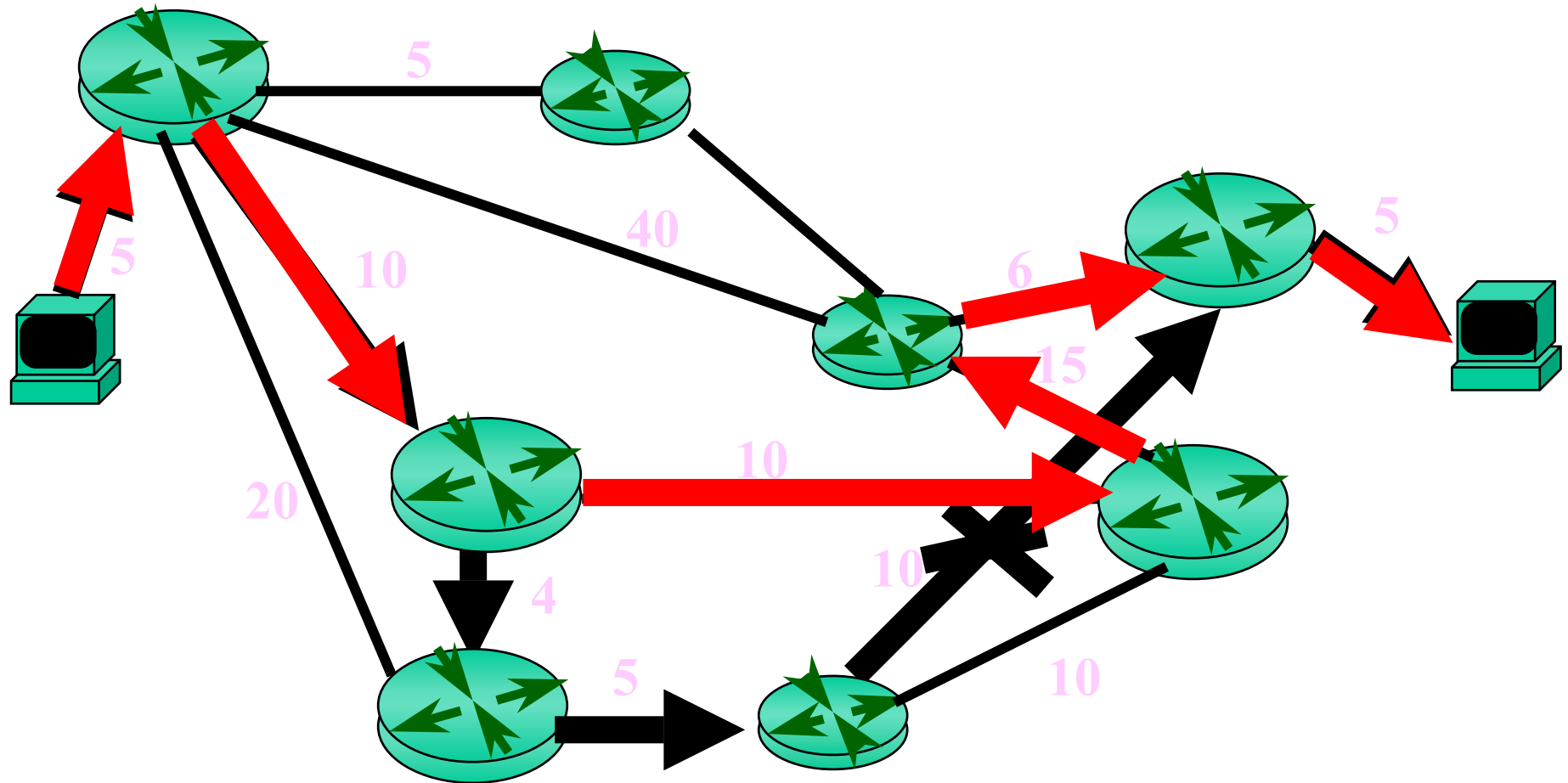


Figure 19.8 Datagram approach

The Datagram Approach



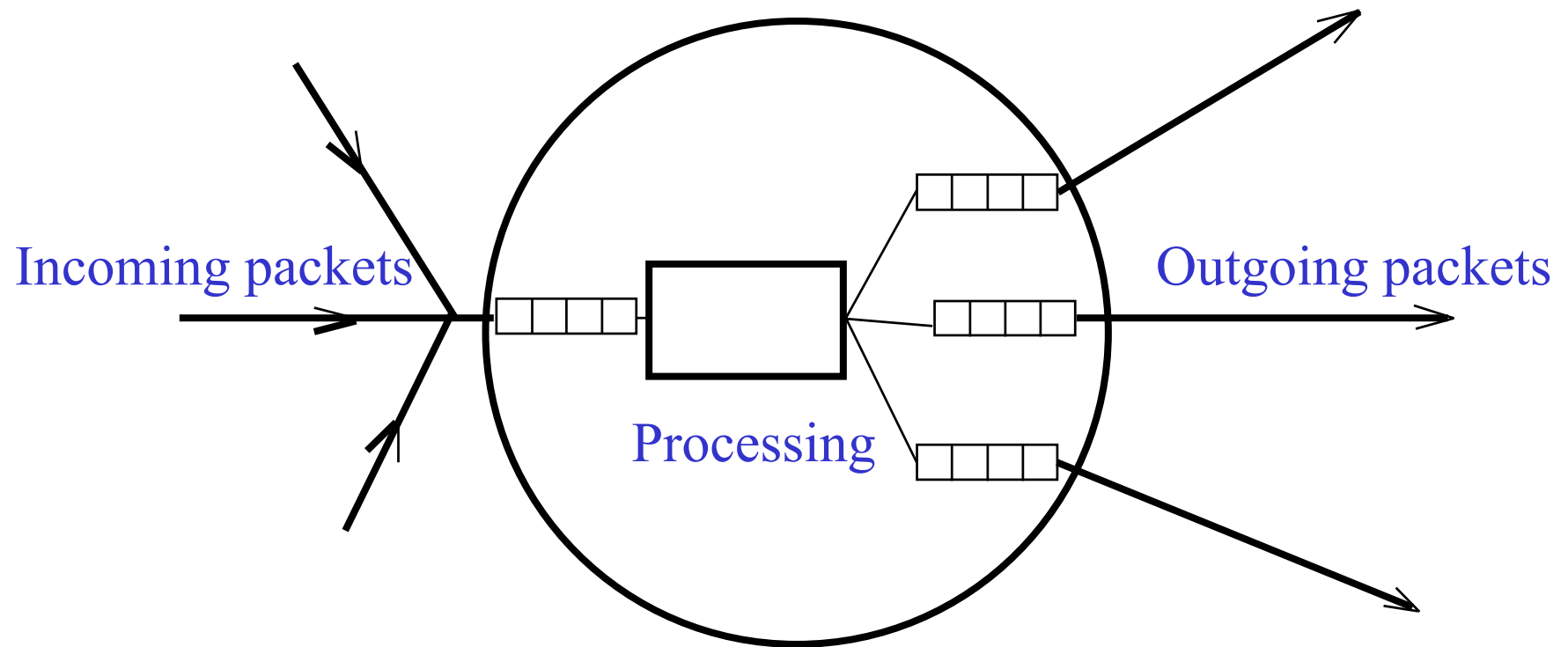
Task: Choosing an Optimal Path



Router

- A router is a type of internetworking device that passes data packets between networks, based on Layer 3 or Network Layer addresses.
- A router has the ability to make intelligent decisions regarding the best path for delivery of data on the network.

Schematic View of a Router



Routing Technologies

- No “intelligence”
- Centralized
- Distributed

Flooding

- Output all packets/datagrams
 - On all ports/interfaces/links
 - Except ingress port/interface/link
- Problem?
 - Looping packets
 - Avoided with TTL counter

Centralized routing technology

- Database and algorithm centralized
 - Network equipment informs central function
- Packet forwarding distributed
 - of course!?

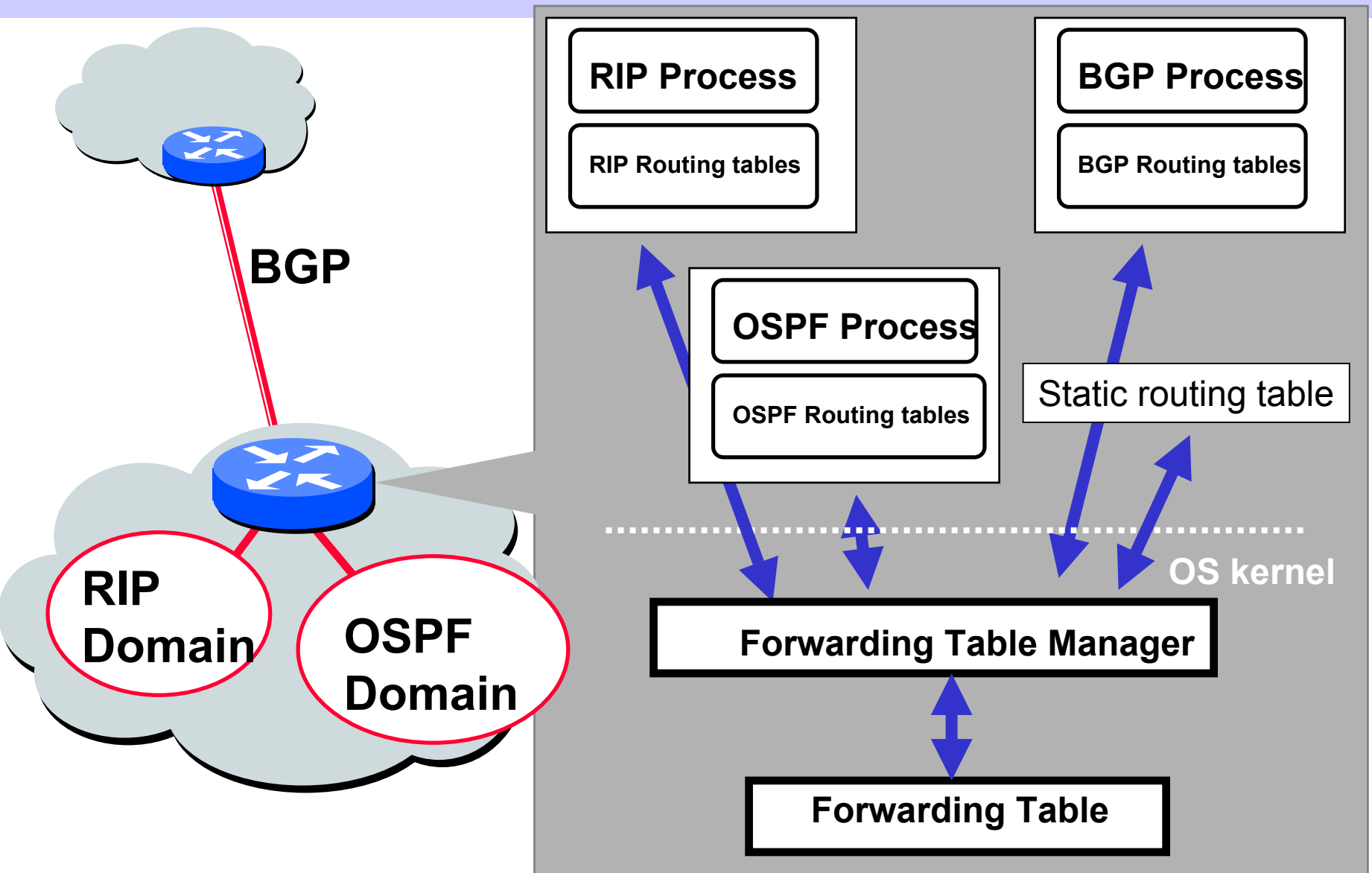
Distributed routing technology

- Routing process distributed among all routers
- Two approaches
 - Distance Vector
 - Each nodes best-path information is distributed to its neighbours
 - Best end-to-end paths result from composition of all next-hop choices
 - Simple, low demands on processor and memory
 - Link State
 - Local topology information is flooded to all nodes
 - Best end-to-end paths are computed locally at each router
 - More complicated, high demands on processor and memory

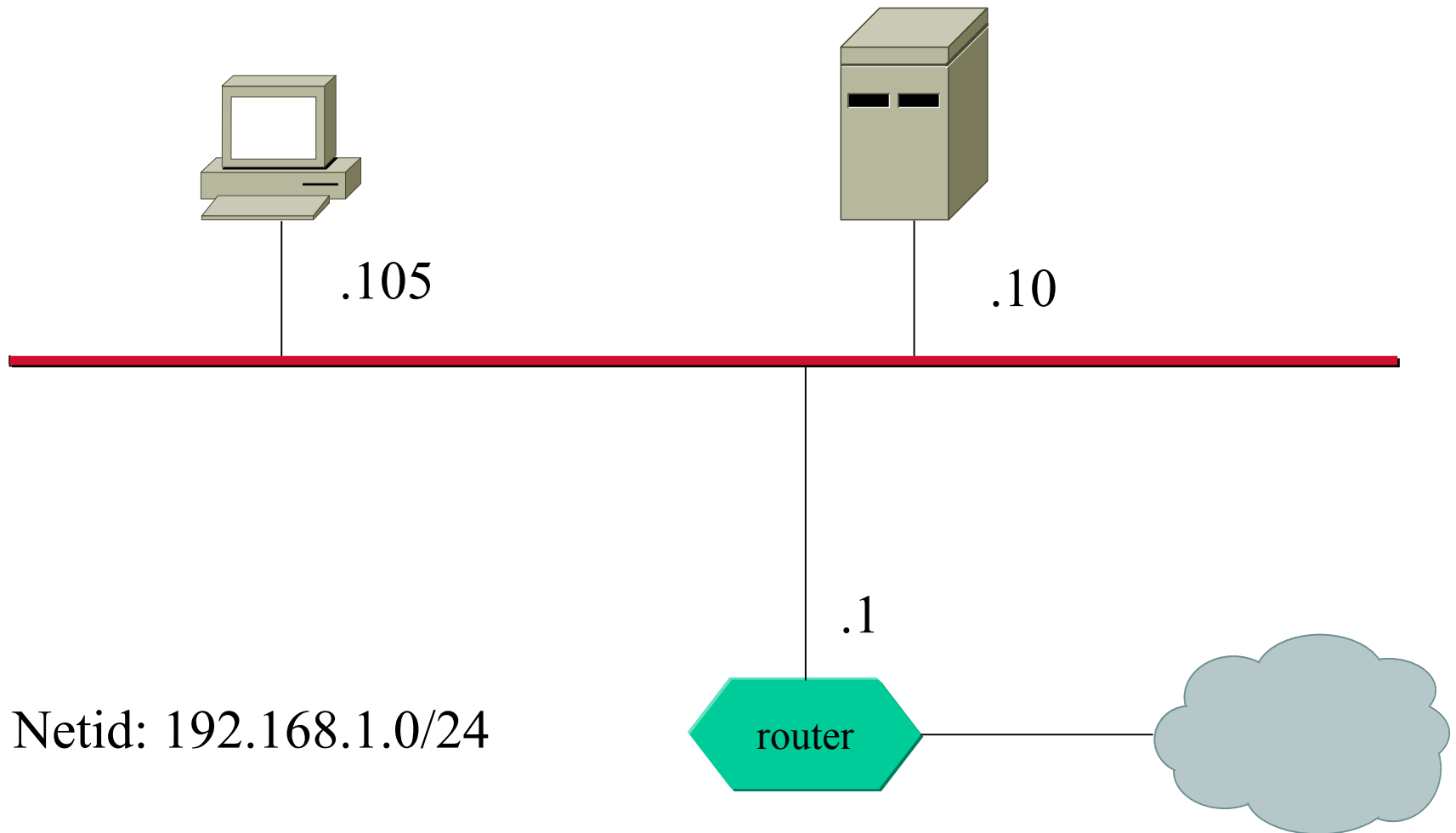
Path Cost, Link Metric

- Possible metrics
 - hop count
 - inverse of the link bandwidth
 - delay
 - dynamically calculated
 - administratively assigned
 - combination

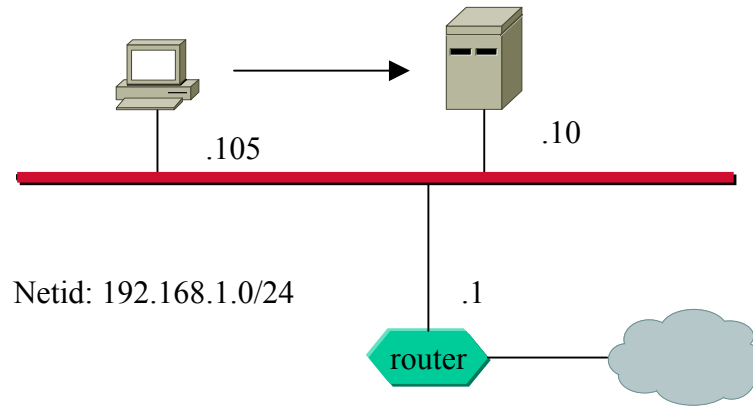
Routing Tables and Forwarding Table



Local Routing & ARP (1)



Local Routing & ARP (2)



Send datagram to 192.168.1.10!

Is destination on same subnet?

Sender compares own net-id with net-id of destination.

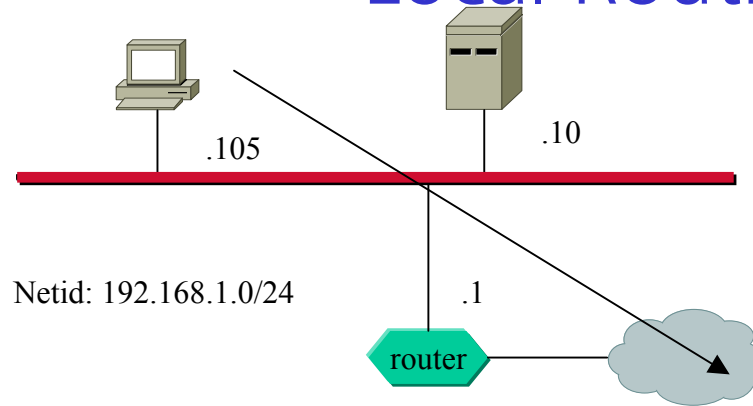
in this case YES

Is MAC-address of destination in ARP-cache?

if YES, use it

if NO, use ARP

Local Routing & ARP (3)



Send datagram to 10.0.100.35!

Is destination on same subnet?

Sender compares own net-id with net-id of destination.

in this case NO

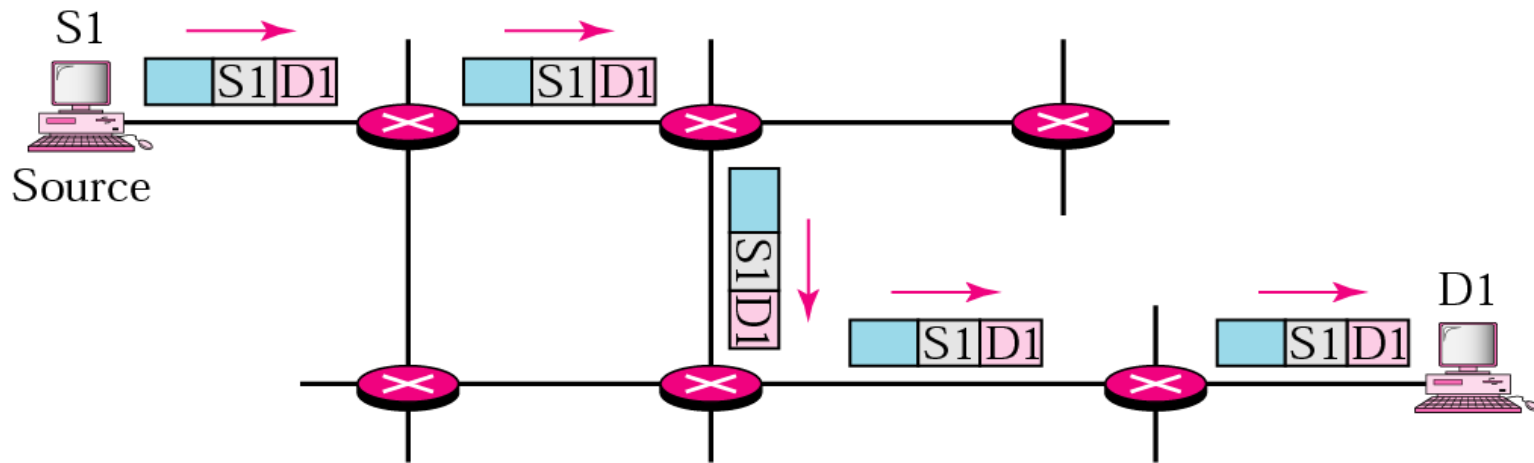
Is network address of default gateway known?

If YES, is MAC-address of default gateway in ARP-cache?

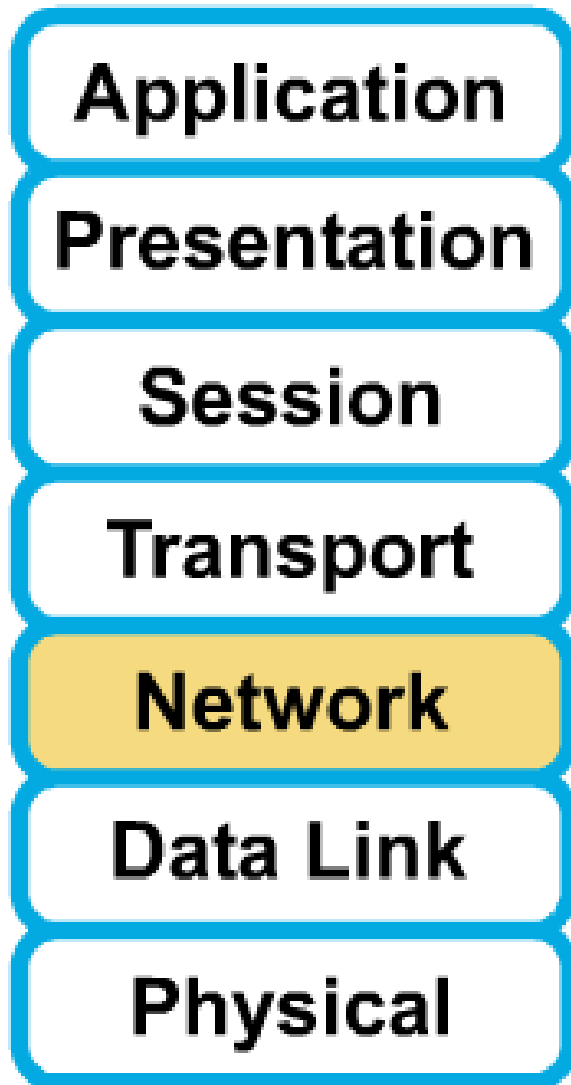
if YES, use it

if NO, use ARP

Unicast Routing



Routable protocols



- Protocols that provide support for the network layer are called **routed** or **routable protocols**.
- IP is a network layer protocol, and because of that, it can be routed over an internetwork.

Routing protocols

Application

Presentation

Session

Transport

Network

Data Link

Physical

- **Routing** protocols determine the paths that routed protocols follow to their destinations.
- Routing protocols enable routers that are connected to create a map, internally, of other routers in the network.

Routing Protocols

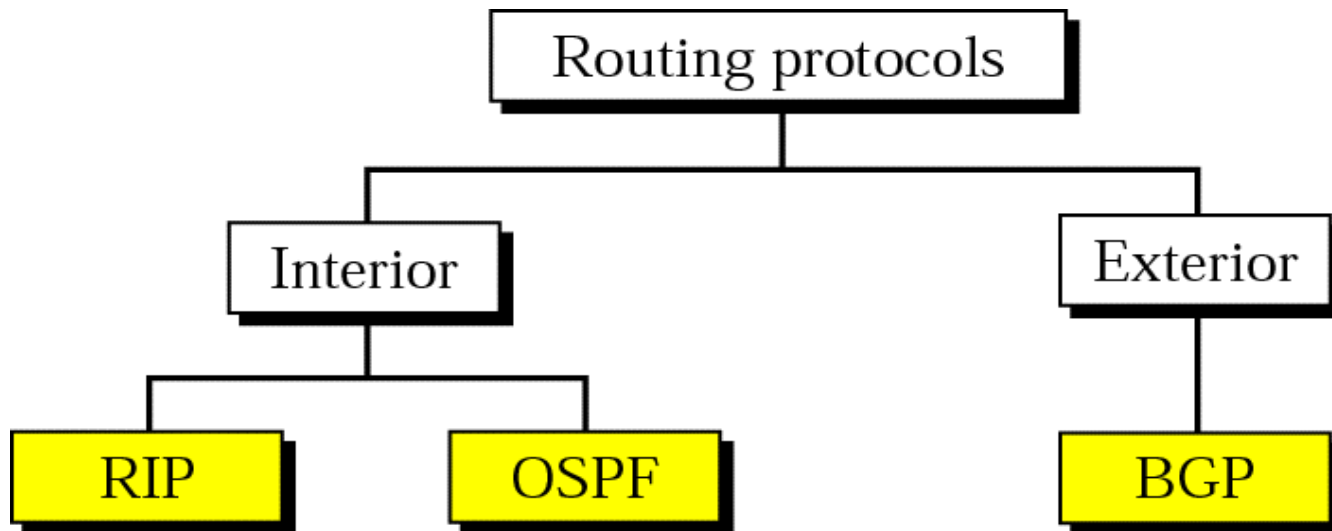


Figure 21-18

Concept of Distance Vector Routing

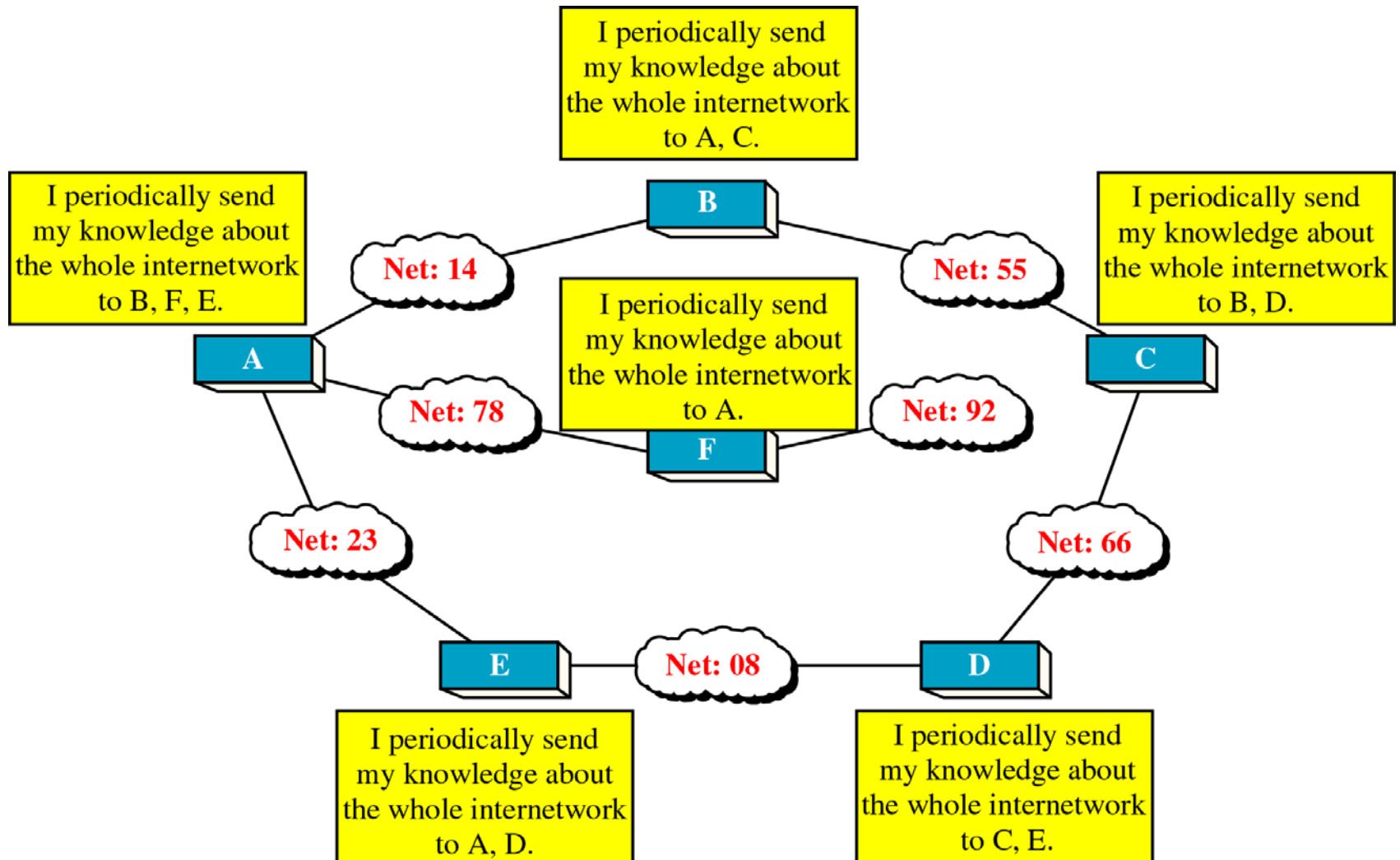


Figure 21-19

Distance Vector Routing Table

Network ID	Cost	Next Hop
• • • • • • • • • •	• • • • • • • •	• • • • • • • • • •
• • • • • • • • • •	• • • • • • • •	• • • • • • • • • •
• • • • • • • • • •	• • • • • • • •	• • • • • • • • • •
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• • • • • • • • • •	• • • • • • • •	• • • • • • • • • •

Figure 21-20

Routing Table Distribution

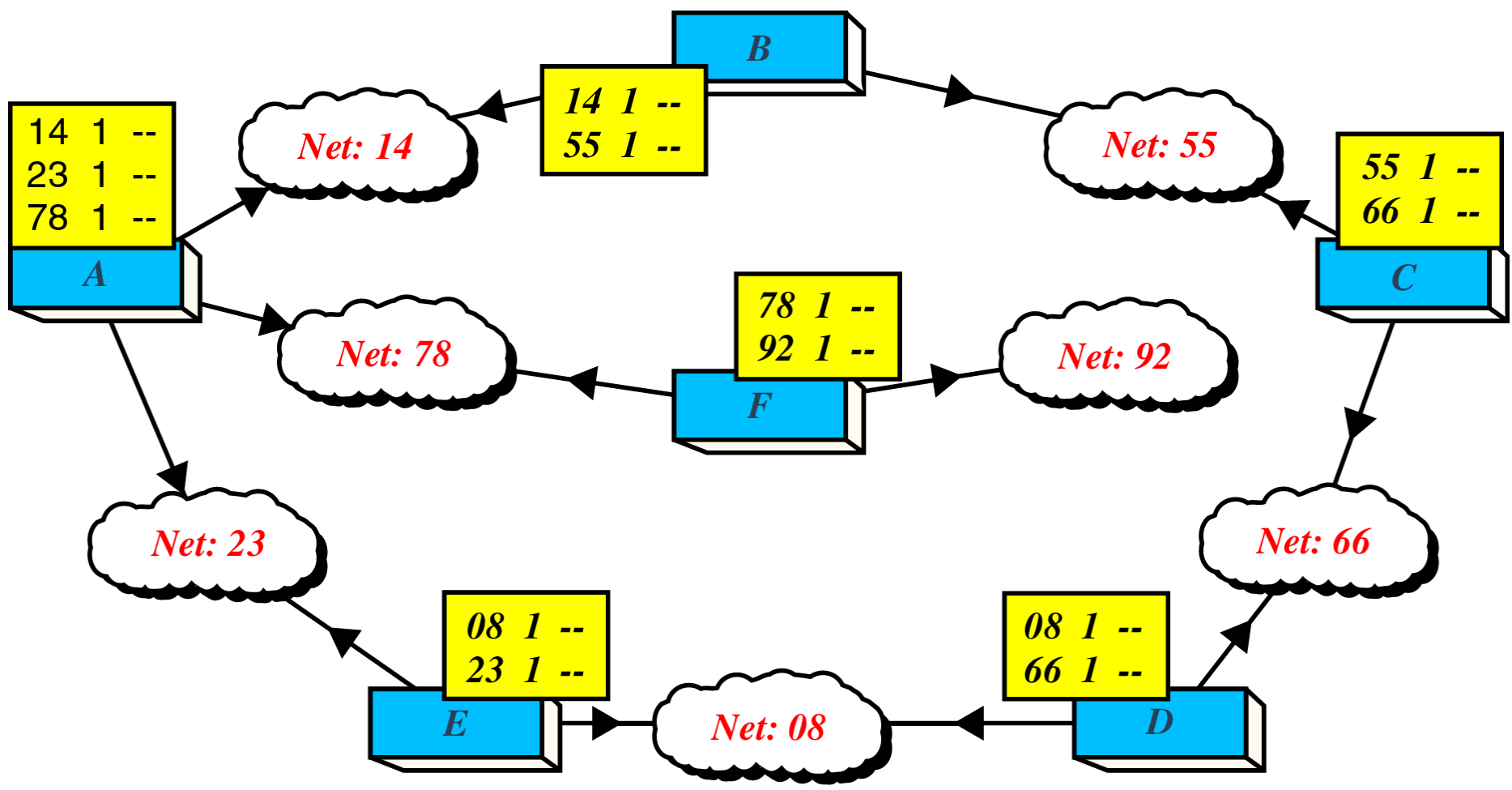


Figure 21-21

Updating Routing Table for Router A

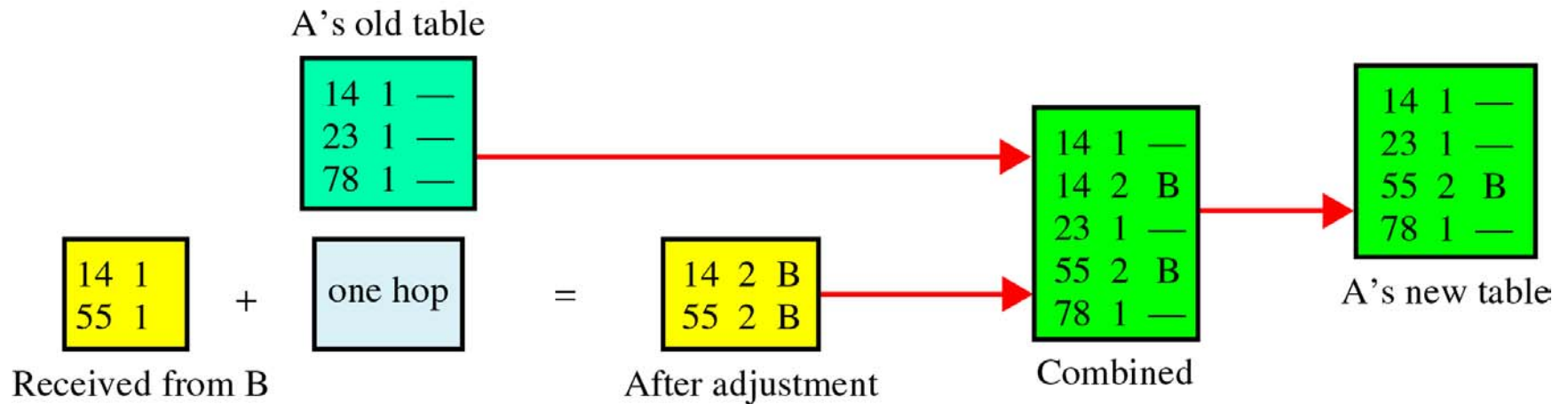
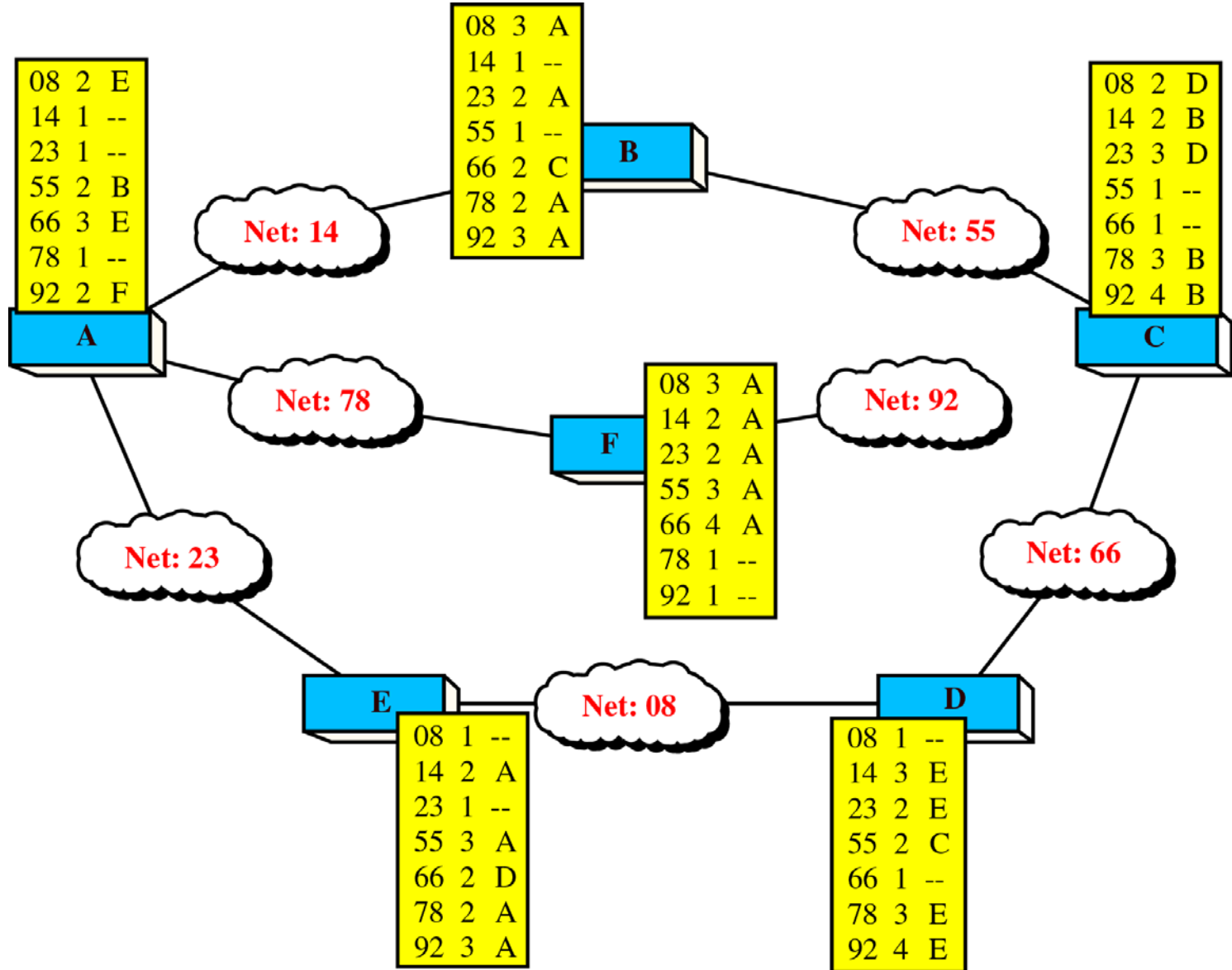


Figure 21-22

Final Routing Tables

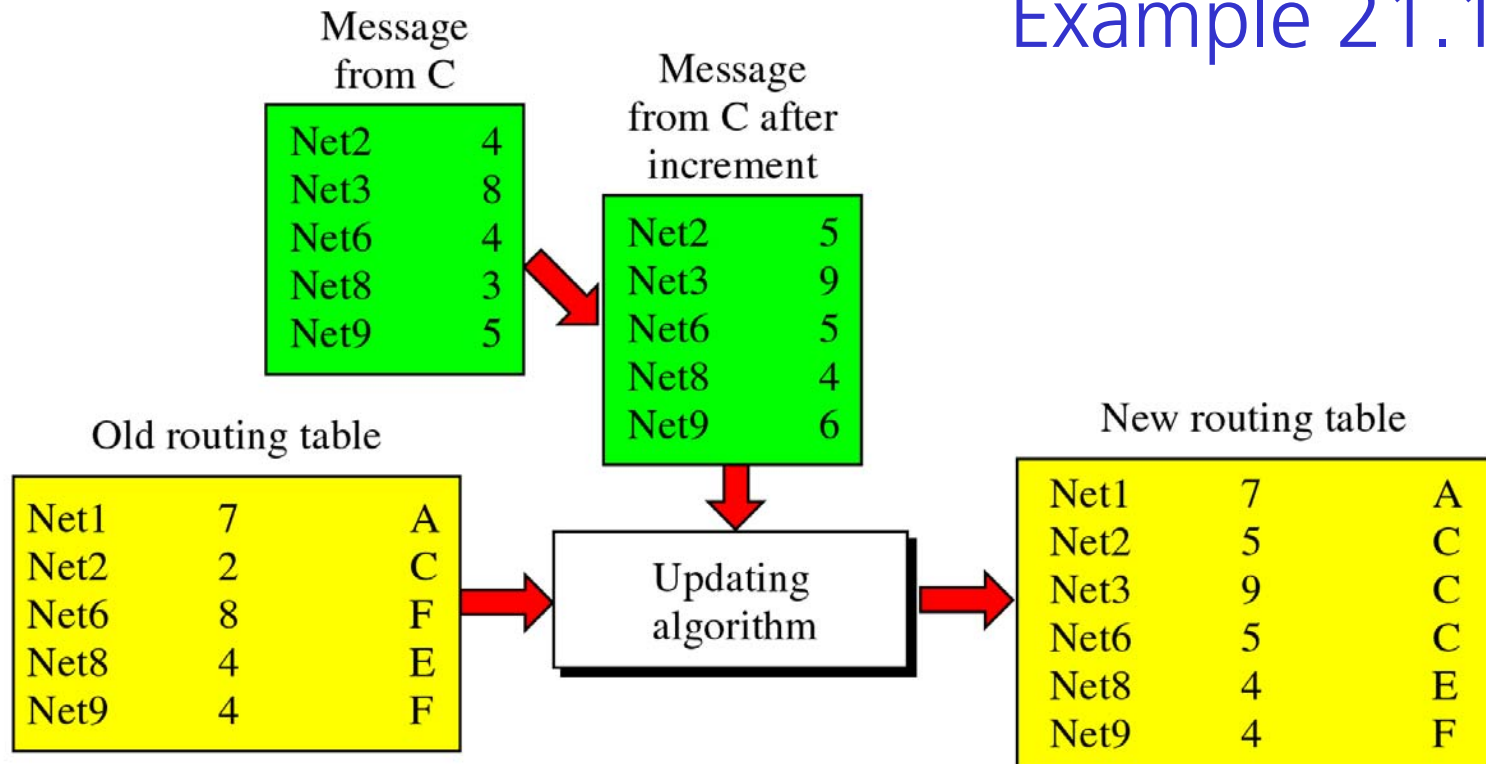


RIP Updating Algorithm

- (1) if (advertised destination not in table) then
 - update table
- (2) else
 - (2.a) if (advertised next-hop = next-hop in table) then
 - replace entry
 - (2.b) else
 - (2.b.i) if (advertised hop count < hop count in table) then
 - replace entry
 - (2.b.ii) else
 - do nothing

Figure 21-23

Example 21.1



Rules

Net2: Replace (Rule 2.a)

Net3: Add (Rule 1)

Net6: Replace (Rule 2.b.i)

Net8: No change (Rule 2.b.ii)

Net9: No change (Rule 2.b.ii)

Note that there is no news about Net1 in the advertised message, so none of the rules apply to this entry.

Figure 21-24

Concept of Link State Routing

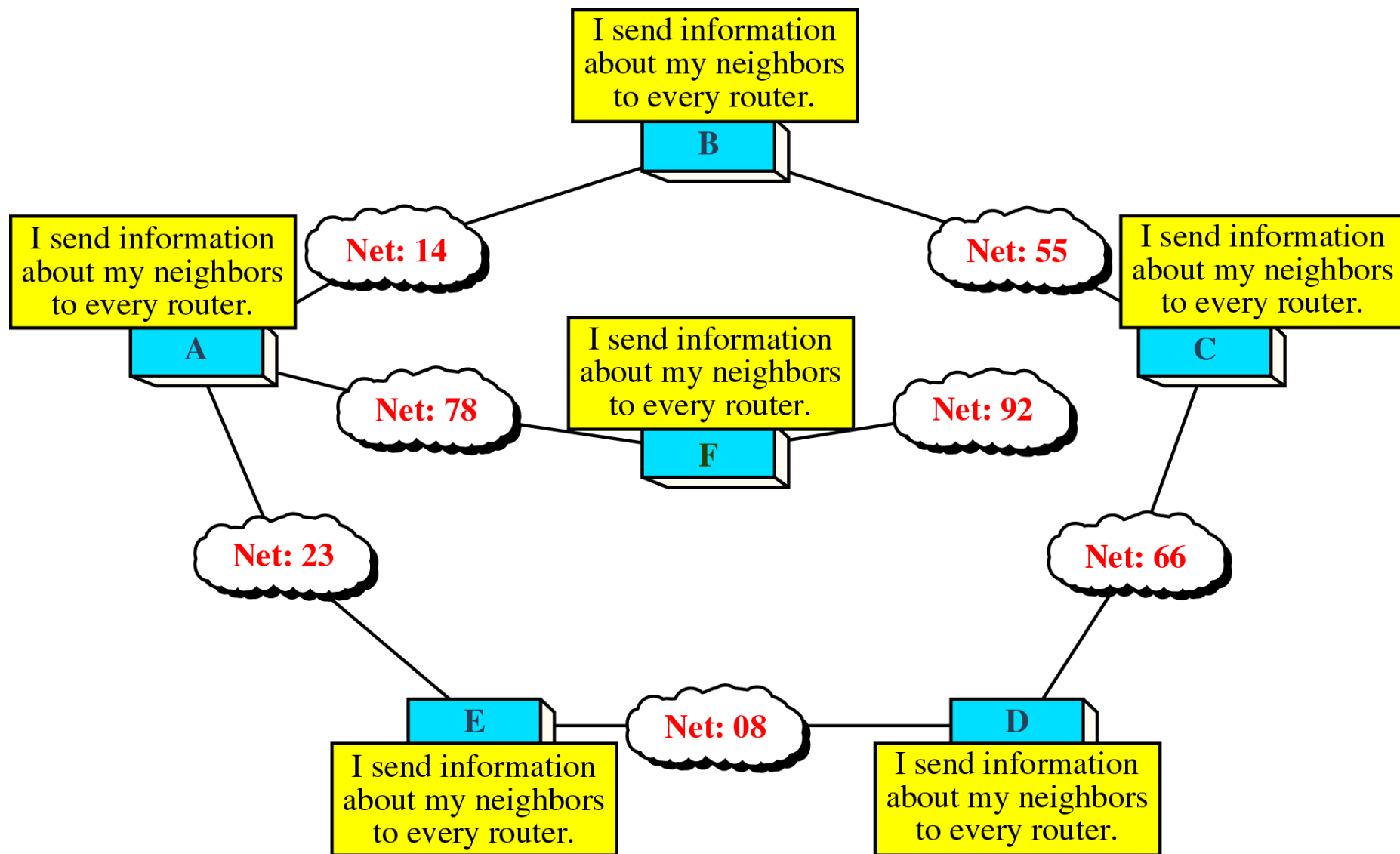


Figure 21-29

Cost in Link State Routing

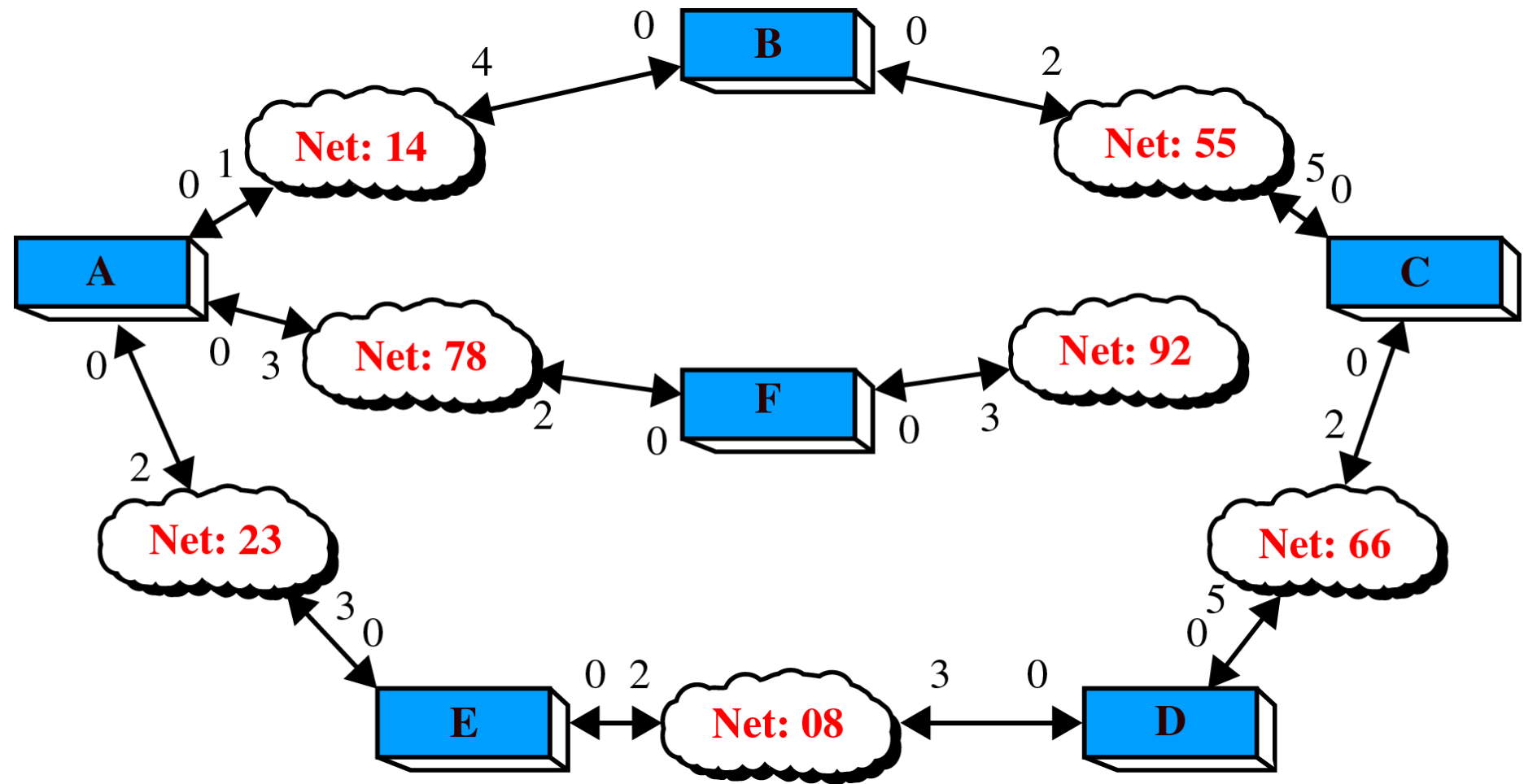


Figure 21-26

Link State Packet

Advertiser	Network	Cost	Neighbor

Figure 21-27

Flooding of A's LSP

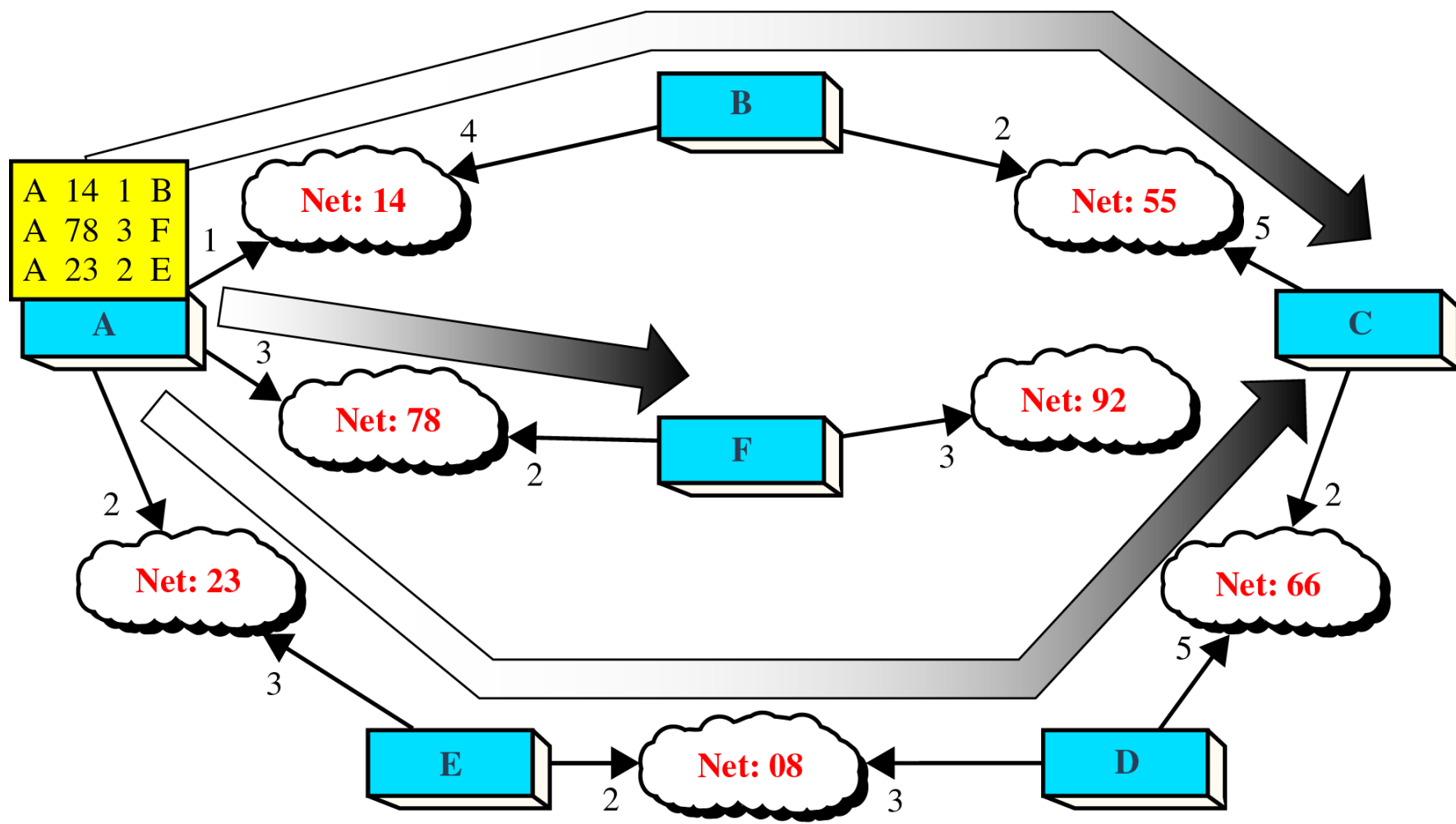


Figure 21-28

Link State Database

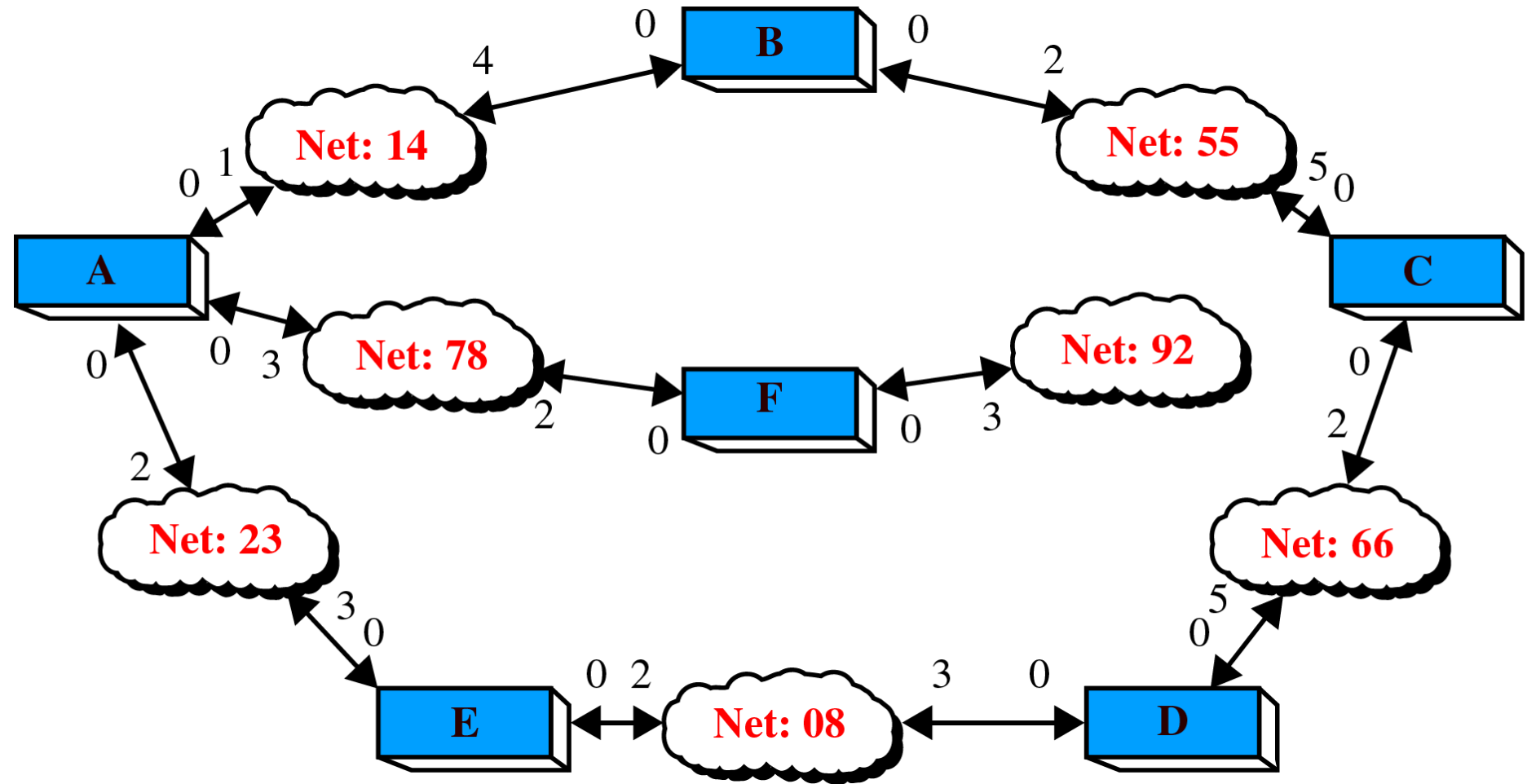
Advertiser	Network	Cost	Neighbor
A	14	1	B
A	78	3	F
A	23	2	E
B	14	4	A
B	55	2	C
C	55	5	B
C	66	2	D
D	66	5	C
D	08	3	E
E	23	3	A
E	08	2	D
F	78	2	A
F	92	3	—

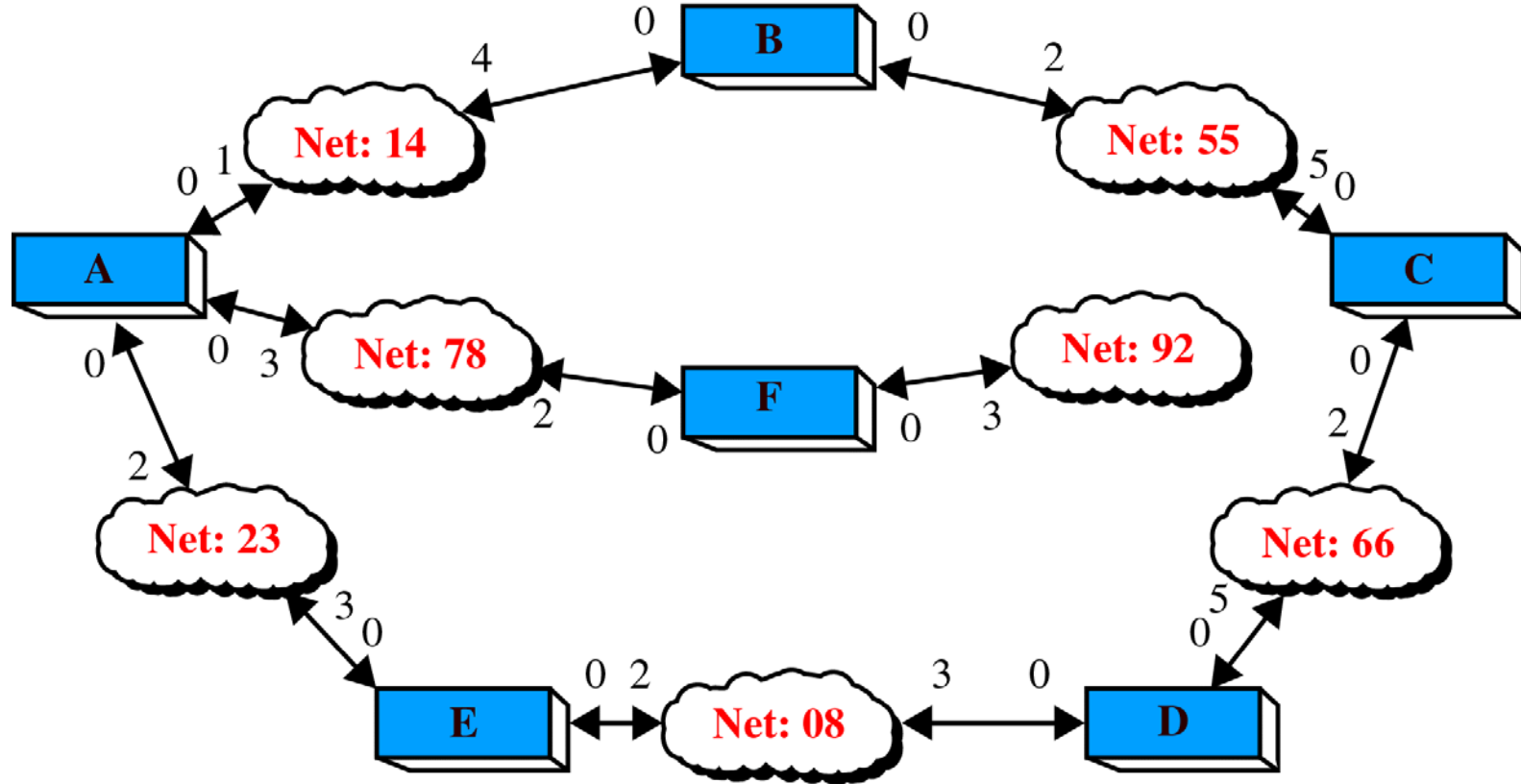
The Dijkstra Algorithm

1. Identify the root (the node itself)
2. Attach all neighbour nodes temporarily
3. Make arc and node with least cumulative cost permanent
4. Choose this node
5. Repeat 2 and 3 until all nodes are permanent

Figure 21-29

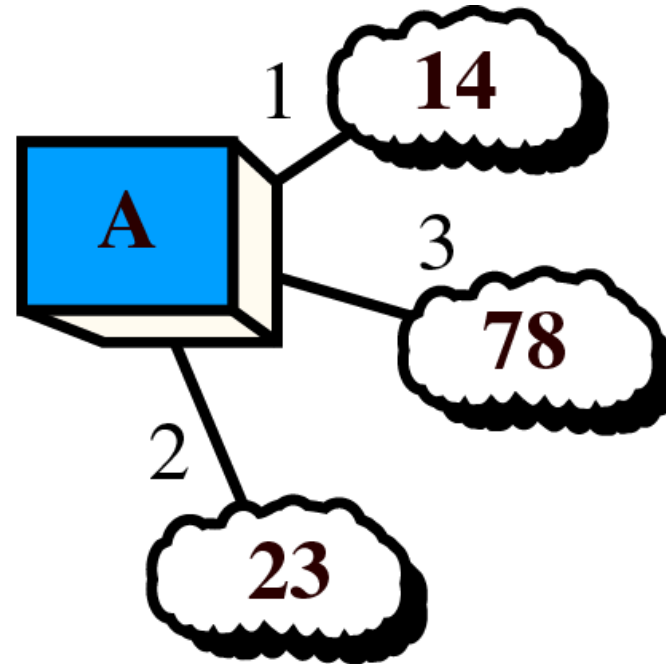
Costs in the Dijkstra Algorithm





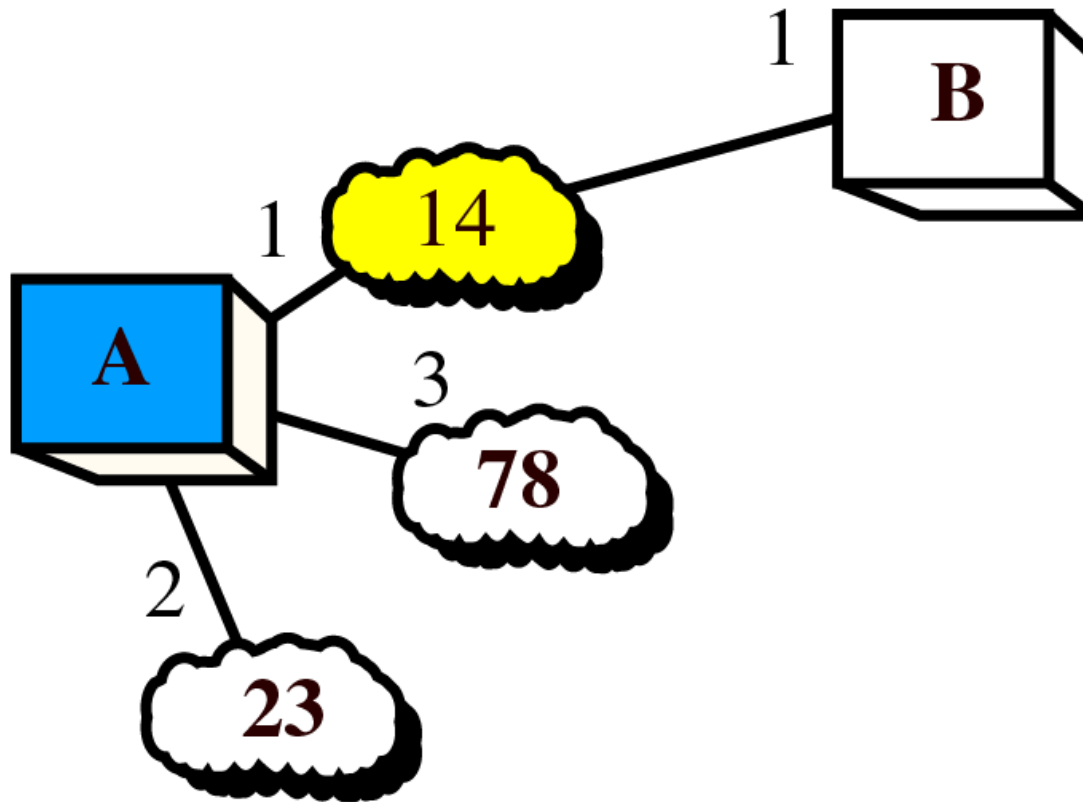
1. Identify the root (the node itself)
2. Attach all neighbour nodes temporarily
3. Make arc and node with least cumulative cost permanent
4. Choose this node
5. Repeat 2 and 3 until all nodes are permanent

Shortest Path Calculation, Part I



Root is A, networks
14, 78, 23 added

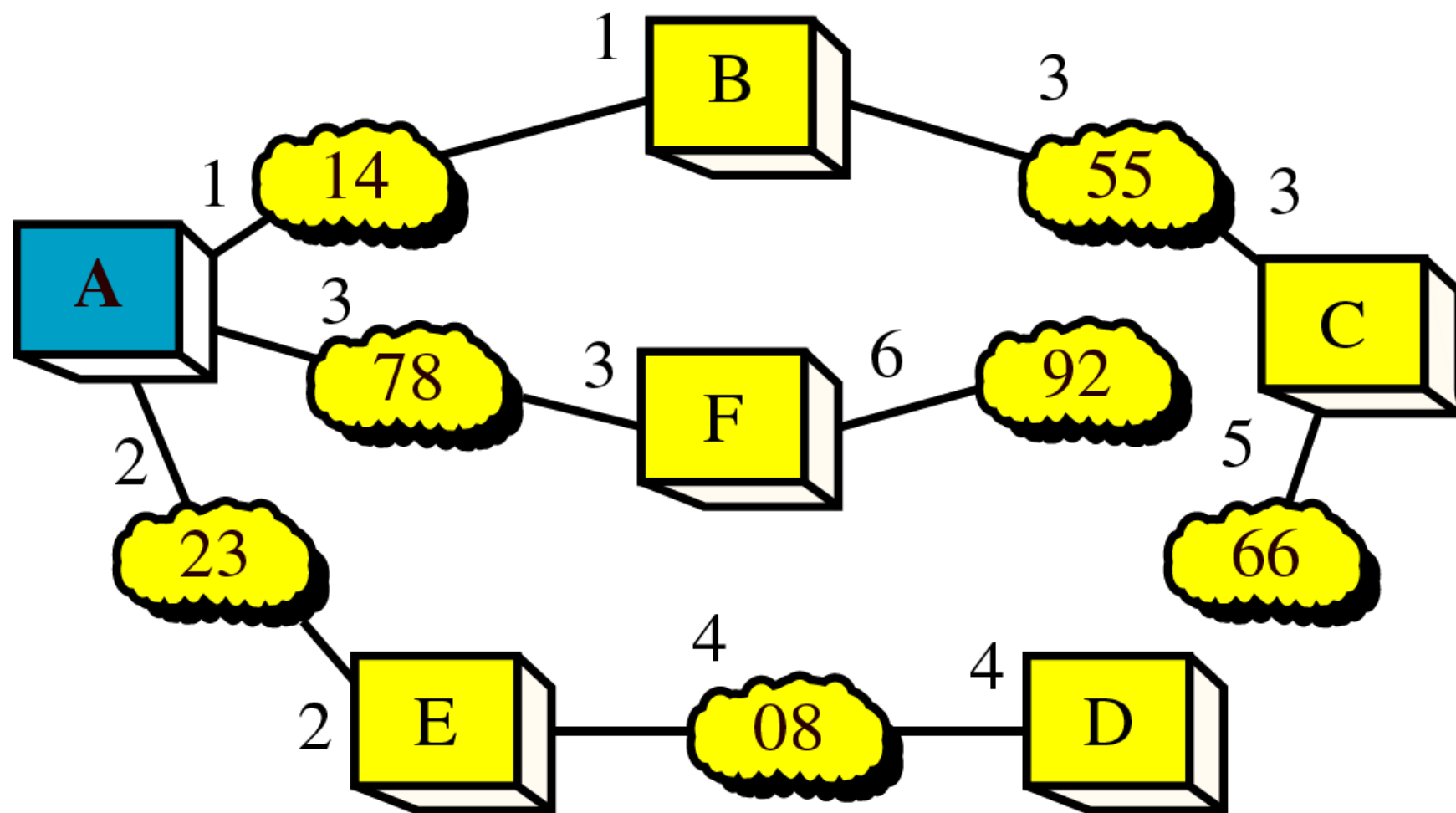
Shortest Path Calculation, Part II



14 permanent, B added

Figure 21-31, Part XIII

Shortest Path Calculation, Part XIII



92 permanent

Figure 21-32

Routing Table for Router A

Net	Cost	Next router
08	4	E
14	1	--
23	2	--
55	3	B
66	5	B
78	3	--
92	6	F

Path Vector Routing

- Distance Vector routing with add on
- Routing Entry
 - Destination
 - Next-Hop
 - Path route has traversed
- BGP

BGP-4

- **BGP** = Border Gateway Protocol
- Is a **Policy-Based** routing protocol
- Is the **de facto EGP** of today's global Internet
- Relatively simple protocol, but configuration is complex and the entire world can see, and be impacted by, your mistakes.

- **1989 : BGP-1 [RFC 1105]**
 - Replacement for EGP (1984, RFC 904)
- **1990 : BGP-2 [RFC 1163]**
- **1991 : BGP-3 [RFC 1267]**
- **1995 : BGP-4 [RFC 1771]**
 - Support for Classless Interdomain Routing (CIDR)

Table 21.3 Path vector routing table

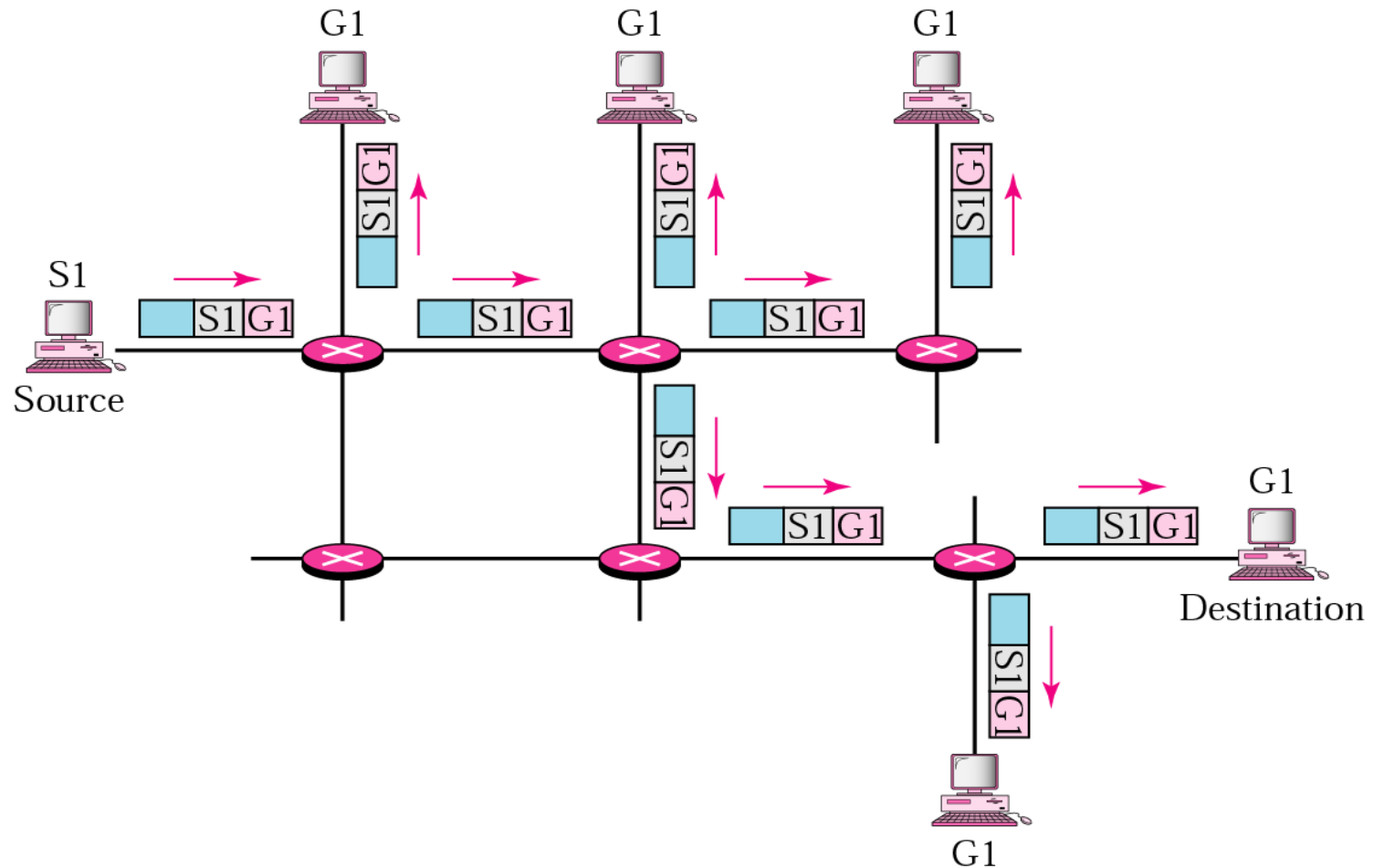
Path Vector Routing Table

Network	Next Router	Path
N01	R01	AS14, AS23, AS67
N02	R05	AS22, AS67, AS05, AS89
N03	R06	AS67, AS89, AS09, AS34
N04	R12	AS62, AS02, AS09

AS = Autonomous System = Organisation

Figure 21.23 Multicasting

Multicasting

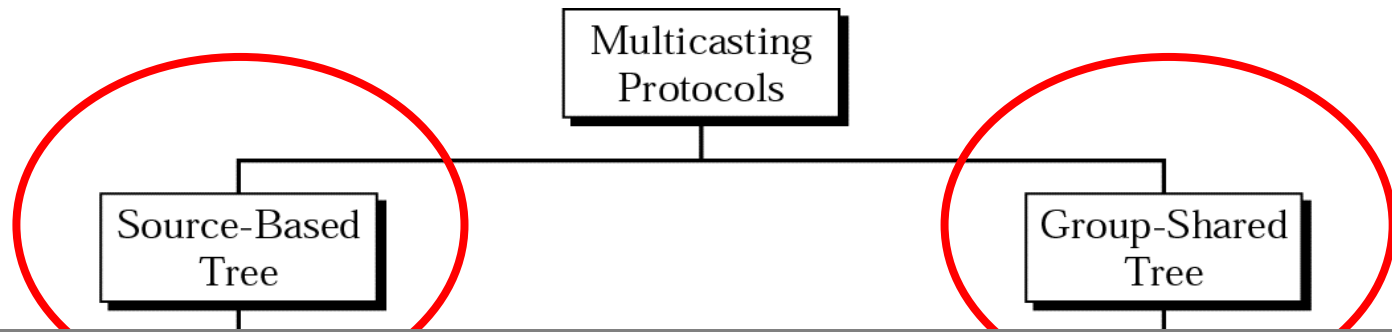


Multicast Trees

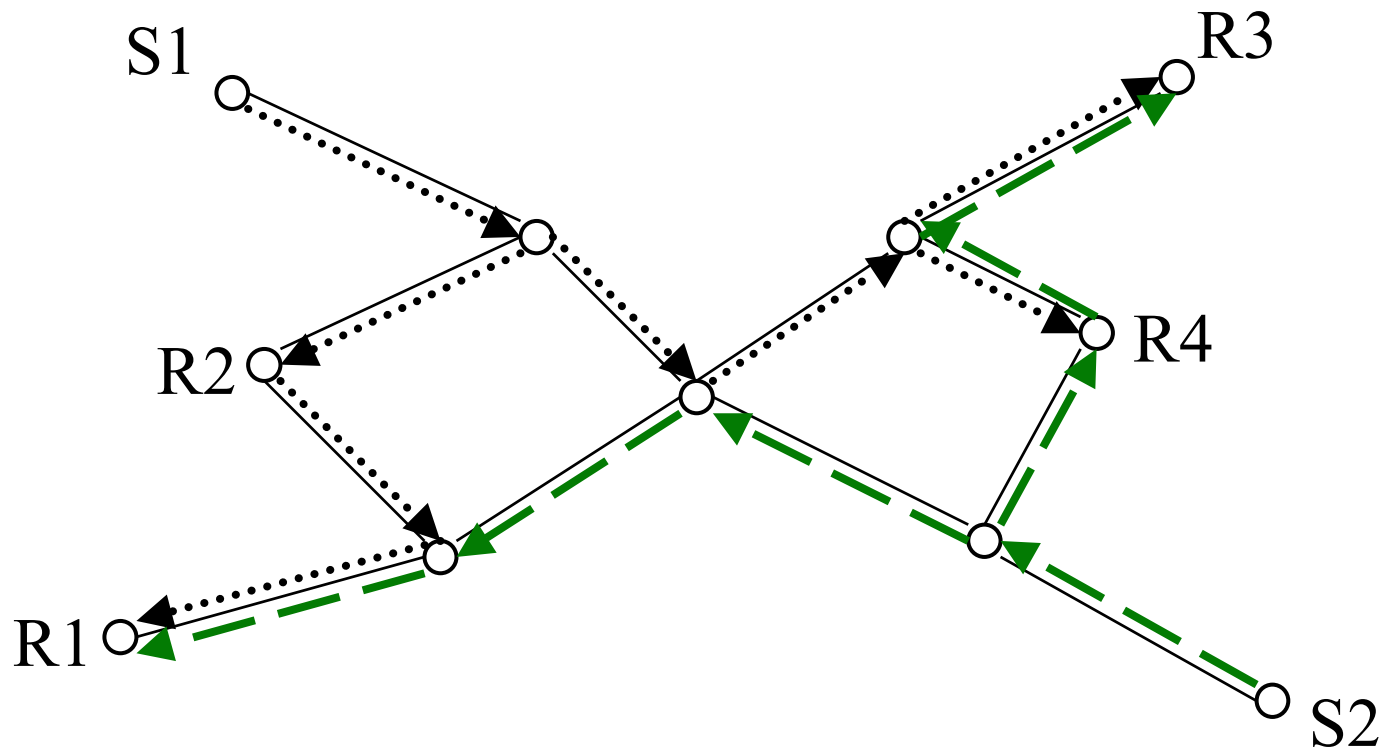
Fundamentals

- Packets address to GROUP
- One and only one copy of each packet to each member
- No packets to nonmembers
- No loops
- Optimal source-member path (shortest path)

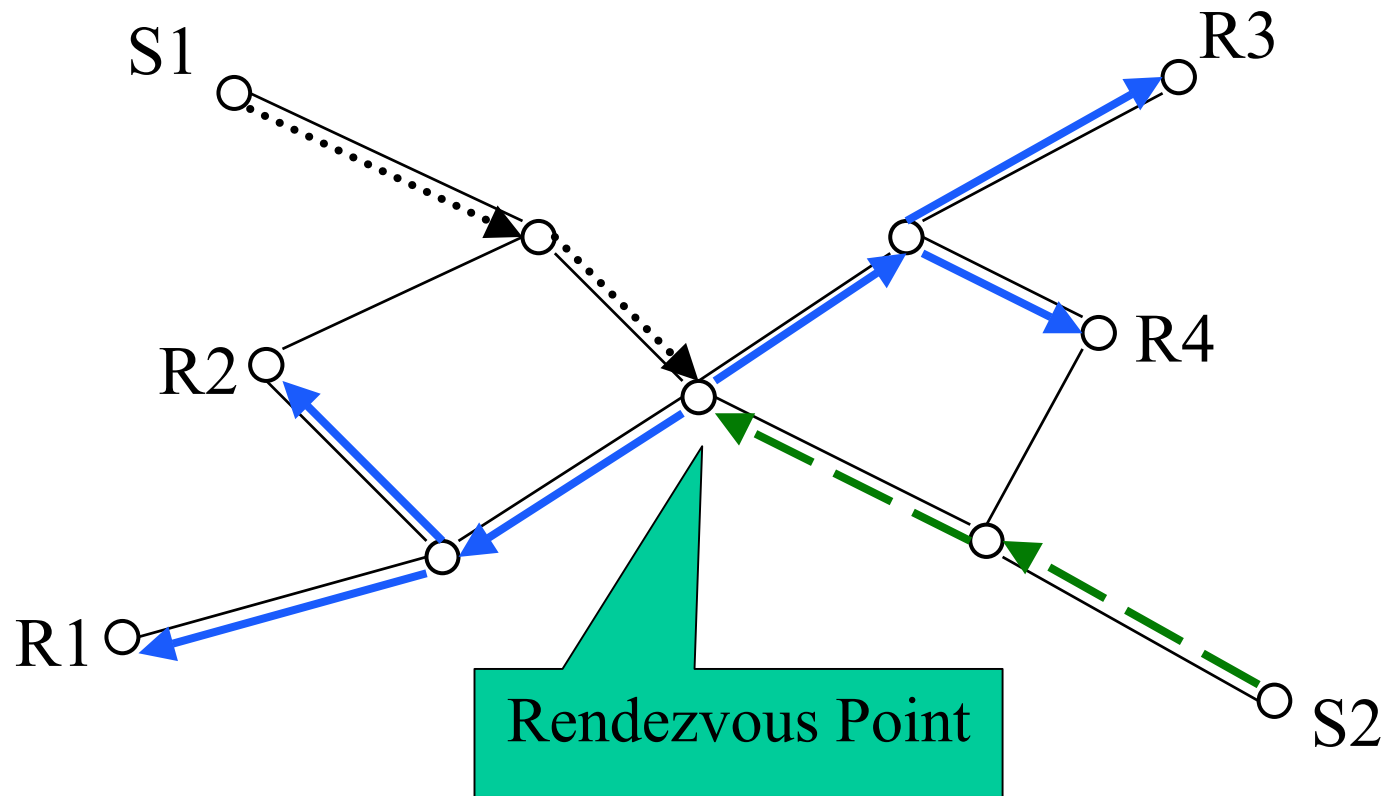
Multicast Routing Protocols



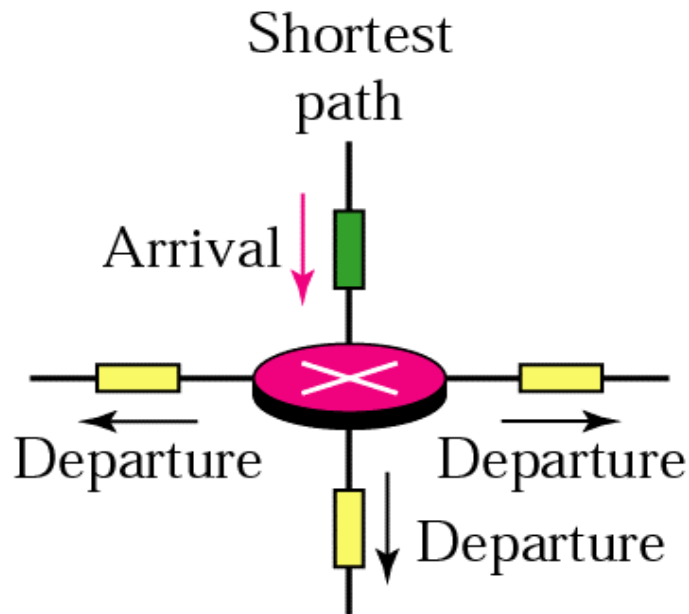
Source-Based Tree, One Tree Per Source



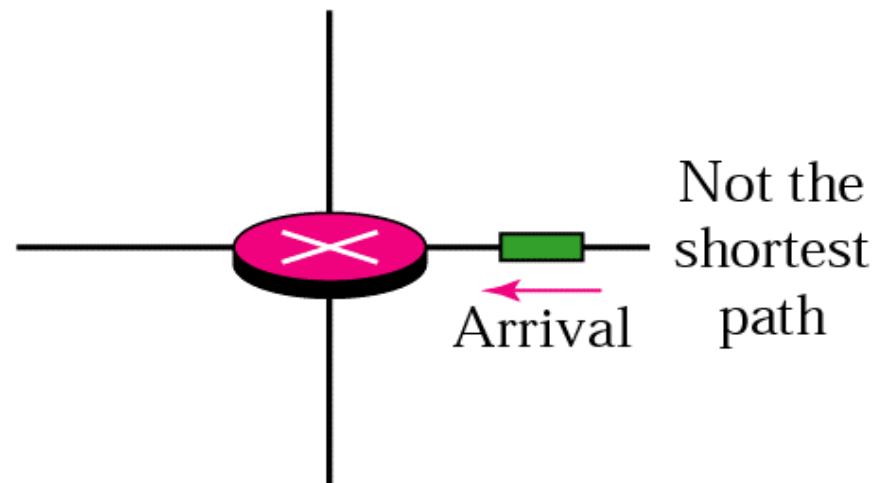
Group Shared Tree, Shared by All Sources



Reverse Path Forwarding



a. Packet is forwarded



b. Packet is discarded